WareVLM: Warehouse Efficient VLM with 600M Parameters

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Introduction

Unlike agent-based systems that rely on tool-calling, our framework achieves superior speed and generalization. We integrate a Geometric Stream to inject 3D centroids, coupled with Weighted Fusion for RGB-D integration. Furthermore, our Answer-Focused Learning strategy ensures the model prioritizes critical spatial keywords.

Methodology

I. Model Architecture

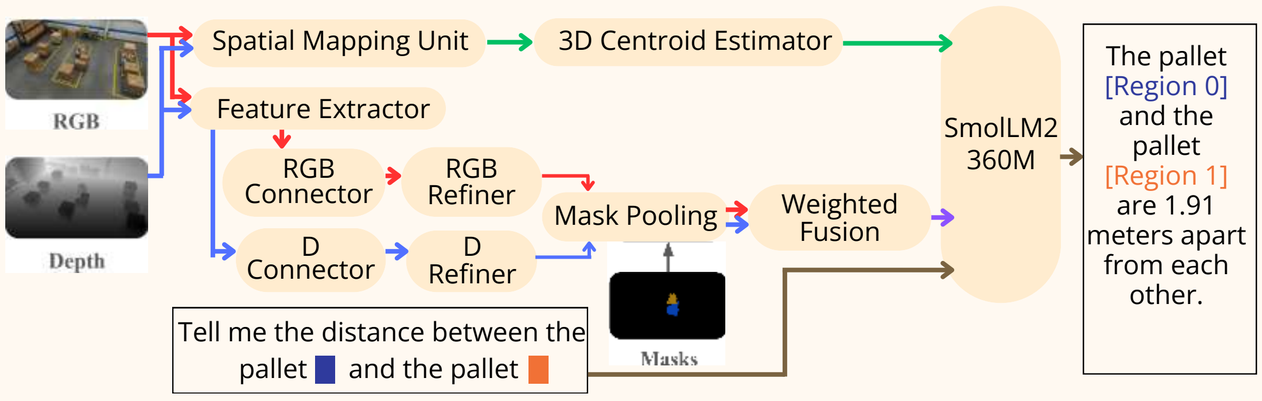


Figure 1. Model Architecture Overview [1.]. The model processes RGB-D inputs through a dual-stream approach. The geometric stream utilizes a Spatial Mapping Unit[2.] and a 3D Centroid Estimator to extract object coordinates, which are injected into SmolLM2. The visual stream employs a Weighted Fusion module[3.] post-mask pooling to balance RGB and depth features before final inference.

II. Evaluation Pipeline

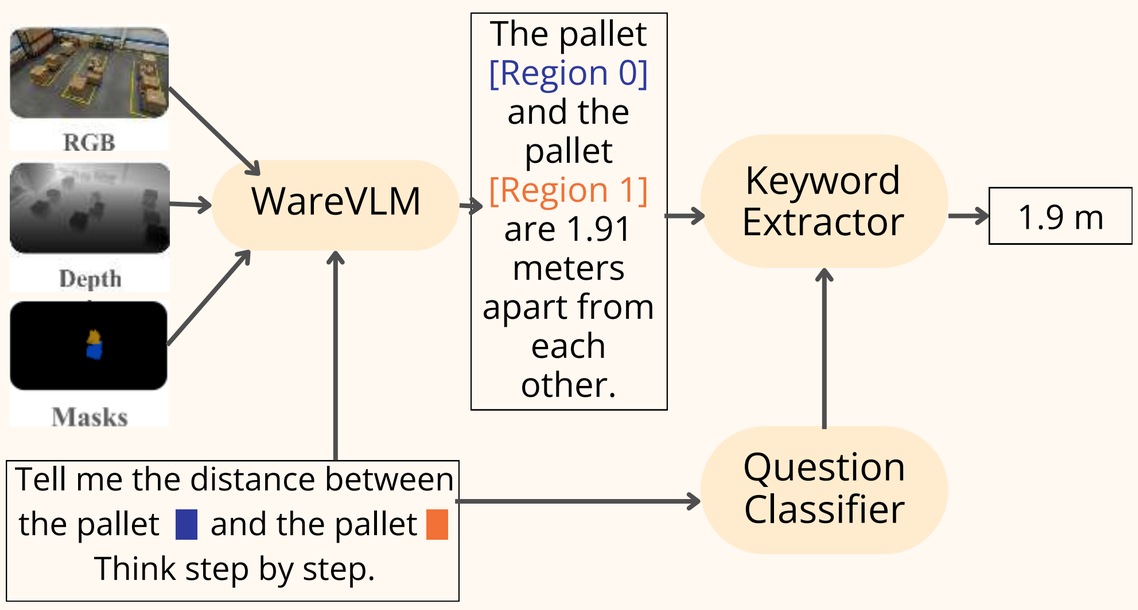


Figure 2. Inference Pipeline with Chain of Thought Prompting

III. Answer-Focused Learning

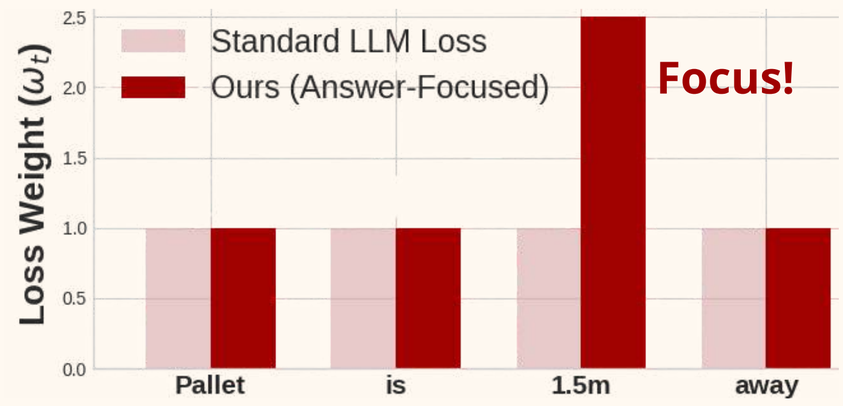


Figure 3. Answer-Focused Learning Method. To ensure the model accurately predicts key answer tokens rather than overfitting to sentence patterns, we introduced an Answer-Focused Learning strategy. We engineered a weighted loss function that prioritizes critical keywords during training.

Reference

[1.] Abdarahmane Traore, Eric Hervet, Andy Couturier. SmolRGPT: Efficient Spatial Reasoning for Warehouse Environments with 600M Parameters. ICCV Workshop 2025.
[2.] Boyuan Chen, Zhuo Xu, Sean Kirmani, Brian Ichter, Danny Driess, Pete Florence, Dorsa Sadigh, Leonidas Guibas, Fei Xia. SpatialVLM: Endowing Vision-Language Models with Spatial Reasoning Capabilities. CVPR 2024.
[3.] Bowen Yin, Xuying Zhang, Zhongyu Li, Li Liu, Ming-Ming Cheng, Qibin Hou. DFormer: Rethinking RGBD Representation Learning for Semantic Segmentation. ICLR 2024.
[4.] Jianwei Yang, Hao Zhang, Feng Li, Xueyan Zou, Chunyuan Li, Jianfeng Gao.
Set-of-Mark Prompting Unleashes Extraordinary Visual Grounding in GPT-4V. 2023.

Results

I. Experiments

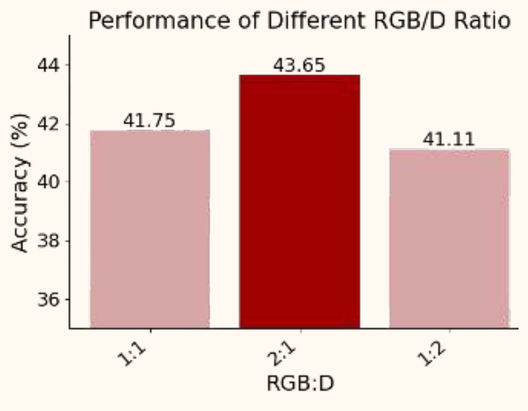


Figure 4. Performance of Different RGB/D Ratio

We investigated the optimal fusion balance between modalities. The results indicate that a 2:1 ratio achieves the highest accuracy (43.65%), supporting the hypothesis that the richer semantic information in RGB images requires higher weighting compared to sparse depth maps.

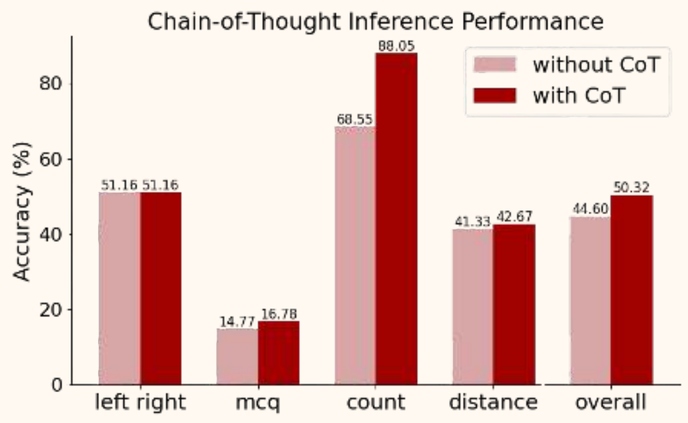


Figure 5. Chain-of-Thought Inference Performance

CoT boosts overall accuracy by 4–7%, with the most significant gain in counting tasks (~20%). However, it provides limited benefit to MCQ/distance and may hinder binary classification (left/right), suggesting that CoT is best suited for complex reasoning rather than simple perception tasks.

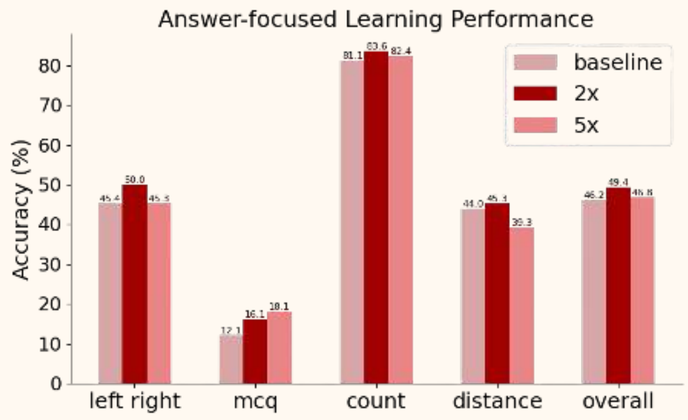


Figure 6. Answer-focused Learning Performance

The 2x setting produces the best results, whereas the baseline and 5x settings have similar overall accuracy. However, performance varies by task: while increasing weights boosts MCQ accuracy, the 5x setting fails in distance estimation, dropping 5% lower than the baseline. This suggests that 2x is the sweet spot—it effectively fine-tunes the model without destroying the pre-trained backbone.

II. Visualization

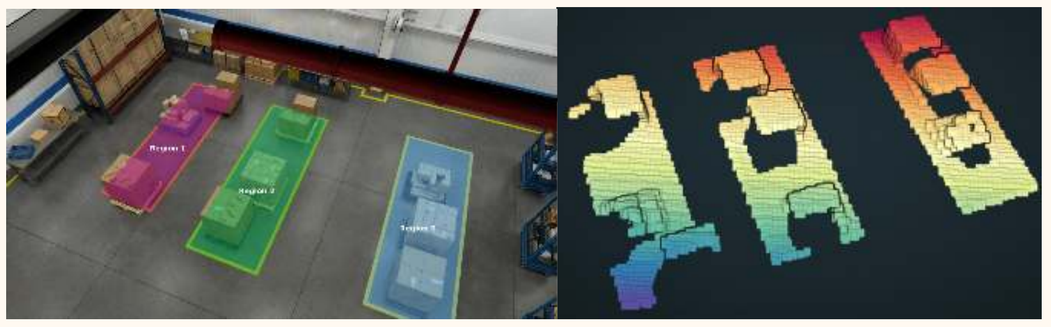


Figure 7. Ground Truth Masks and Point Cloud for Regions



Figure 8. Ground Truth Masks and Point Cloud for Pallets

Future Work

1. Visual Grounding with Set-of-Mark: Mitigating hallucinations and resolving referential ambiguities.[4.]
2. Advanced 3D Integration: Exploiting depth features via cross-modal attention for finer geometric understanding.